

Chorus

A bucket-brigade stereo chorus in the spirit of the classic 80s poly-synth choruses, with modes I, II and I+II.



Overview

Chorus mixes your signal with two short delayed copies, one per side, whose delay times are swept by a slow triangle LFO. The right side sweeps in the opposite direction to the left, so the image spreads wide while the mono sum stays solid - the sound of the built-in chorus on the classic 80s poly-synths.

The delay lines behave like bucket-brigade chips: the signal is filtered before and after the delay, gently compressed and saturated, and can carry a little hiss. TONE sets how dark the repeats are.

Three mode keys recall the classic settings: I is slow and gentle, II faster and deeper, and I+II adds a second, faster sweep running the other way for a richer, wider sound without the throb of a single fast sweep. From there every knob can be adjusted.

Reach for it for

- Width and shimmer on synth pads and strings
- Clean guitars that need movement
- Thickening a mono source into stereo
- Vintage, slightly dark chorus with hiss for character
- Tempo-synced swirls on loops

Controls

Modes (header)

I	Slow and gentle: 0.513 Hz, 1.85 ms depth around a 3.5 ms centre.
II	Faster and deeper: 0.863 Hz with the same depth and centre.
I+II	Both at once: the I sweep (set by the knobs) plus a second tap swept about 1.68 times faster, a little longer and swinging the other way. Richer and wider.
OFF	The chorus is bypassed; the IN / OUT stage and OUTPUT stay active.

Pressing I, II or I+II sets RATE, DEPTH and CENTRE to the classic values for that mode and the rate mode to FREE. The red LEDs above the keys show which sweeps are running.

Display

Delay window	The two dots show the live delay time of the left (cyan) and right (orange) taps on a 0 - 10 ms scale, with the times in ms above. CHORUS OFF is shown when the chorus is bypassed.
LFO	The triangle sweep with a dot riding on it.

Chorus

FREE / SYNC / FLOW (RATE MODE)	How the sweep speed is set: FREE in Hz, SYNC locked to the project tempo and timeline, FLOW tempo-synced but running continuously through loops. In I+II the second sweep runs freely at its scaled speed when synced.
RATE	The sweep speed, 0.05 to 12 Hz in FREE, or a note value from 4 bars to 1/32 (dotted and triplet included) when synced.

DEPTH	How far the delay time swings either side of the centre, 0 to 3 ms. More depth gives more pitch wobble.
CENTRE	The delay time the sweep moves around, 1 to 8 ms. Short settings are tighter; longer ones sound bigger and more doubled.
WIDTH	0 to 100%. At 100% the right side sweeps in antiphase to the left for the classic wide image; at 0% both sides sweep together (a mono-style chorus).
TONE	The bucket-brigade filtering on the delayed signal, 0 to 100%: low settings roll the repeats off around 3 kHz for a dark, vintage sound; high settings open them up towards 16 kHz.
HISS	Adds the faint noise of the delay chips into the chorus path, 0 to 100%. Off by default.

Drift

A slow random wander of the chorus rate, its depth and its tone, like ageing components.

DRIFT	0 to 100%. 50% gives the plugin's standard drift (the amounts the presets set), 100% doubles it, 0 turns it off. The wander also speeds up as you turn it up.
--------------	---

IN / OUT

TRIM	Input level before anything else, from -24 to +12 dB.
GUARD	On by default. Click the TRIM knob (without dragging) to switch it; the small LED at the knob's upper right shows it is on and glows brighter while it works. It gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources.
SAT	Band saturation on the way out, 0 to 100%: only the lows and highs are saturated (the mids stay clean), with a soft

	curve that loosens as the level rises.
OFF / CLIP / LIMIT (OUT)	The final stage: nothing, a soft clipper (no latency) that holds the peaks at -0.3 dBFS, or a lookahead limiter (3 ms latency, compensated by REAPER) that holds them at -0.3 dBFS. The CLIP key flashes red while the clipper is working.

Master

MIX	Dry / chorus balance, 0 to 100%; 50% (the default) is the classic even blend.
LOW CUT	A 12 dB/oct high-pass on the chorused signal only, from off (fully left, 20 Hz) up to 500 Hz. The bass stays dry, clean and centred at full level.
OUTPUT	Output level, -24 to +12 dB.
OUTPUT LEVEL meters	Output level, left and right. Click to reset the peak hold.

The header

PRESETS	Opens the list of factory presets; the arrows beside it step through them, and the display shows the one in use (click it for the list too). The last entry resets everything to the defaults.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Presets

The presets set the mode, rate, depth, centre, width, tone, hiss, mix and drift amounts; the DRIFT dial, LOW CUT, OUTPUT and the IN / OUT stage are left alone.

Classic I	Mode I at its classic settings.
Classic II	Mode II at its classic settings.
Classic I+II	Both sweeps together at the classic settings.
Lush Wide	Deeper (2.5 ms) and longer (5 ms centre) sweep at 0.6 Hz, a little more mix and drift.
Subtle Thicken	Slow 0.3 Hz, shallow 0.8 ms sweep, 70% width and 30% mix for gentle thickening.
Vintage Hiss	Mode II with a darker tone and audible chip hiss.
Synced Swirl	Mode II locked to half-note sweeps of the project tempo, a little deeper and longer.
Dark Bucket	Slow sweep with the tone filter well down, a touch of hiss and more tone drift.
Reset all to defaults	Every control back to its starting value.

Tips

- On bass or kick-heavy material, raise LOW CUT to 150 - 250 Hz so the low end stays solid and mono.
- I+II is the one to try when II sounds too seasick: it is just as wide, but the throb is smoother.
- For a mono-compatible effect, pull WIDTH down a little; at 100% the antiphase sweep is at its widest.
- Lower TONE and add a little HISS for the lo-fi sound of an old synth's chorus.

File: gfx-chorus.jsfx